

Compact Note Environment Options

ECMA 30760 sample renderings

Spring 2026

Goal	Preserve the current lecture-note structure, including global contents, chapter sub-contents, margin notes, and colored theorem-like environments.
Palette	Same color families as the existing notes: green, pale green, yellow, pink, and brown.
Flex points	Colored area, title placement, box padding, rule weight, title/body spacing, and how much of the environment name is visually separated from the text.

Option 1: Tight Attached Tags

Closest to the current design, with less padding and a smaller attached title tag. It keeps the colored body as a soft field while reducing vertical overhead.

Template elements preserved

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Voting	
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1.2 Gibbard–Satterthwaite	7

1.1 Introduction

Formally, $\succ_i$ is a *binary relation*; i.e., it is a subset of $A \times A$ with the infix notation of $x \succ_i y$ instead of $(x, y) \in \succ_i$.

Importantly, i does not depend on $\succ$. The dictatorial property says that i 's top choice is chosen for all preference profiles.

Definition 1.1.3 (Social choice function).

A social choice function is a mapping $f : P^N \rightarrow A$.

Informally, a social choice function is a decision rule for each preference profile.

Example 1.1.5.

For instance, if $A = \{w, x, y, z\}$, the plurality rule chooses w , instant runoff chooses x , elimination chooses y , the Borda count chooses x , and the Copeland rule chooses x .

Lemma 1.1.9.

If X and Y are dominating sets, then $X \subseteq Y$ or $Y \subseteq X$.

Proof of Lemma 1.1.9. Suppose not. Then any $x \in X \setminus Y$ and $y \in Y \setminus X$ have to both beat each other in head-to-head elections, which cannot happen under strict preferences. $\square$

Theorem 1.2.5 (Gibbard–Satterthwaite).

If $|A| > 2$, then every surjective and strategy-proof social choice function is dictatorial, in that there is some agent i such that

$$f(\succ) = \arg \max_{\succ_i} A$$

for all preference profiles $\succ$.

Option 2: Full Header Bands

The color mostly lives in the header band. The body is nearly white, so long definitions use less visual weight while remaining easy to scan.

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Option 3: Left Color Rail

The colored area is reduced to a vertical rail plus a faint body tint. Environment names sit close to the body text, making this the quietest boxed version.

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Option 4: Inset Inline Tags

A small colored tag is inset into a hairline box. This compresses the title/body gap while retaining an explicit box boundary.

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Option 5: Run-In Capsules

The environment label becomes a compact capsule at the start of the paragraph, so the body begins on the same line when space allows.

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